

Testing of Hypothesis: Proportion

Reading Objectives:

In this reading, you will learn about different models related to testing hypotheses concerning proportion. You will also learn about different models related to proportions depending on the given hypothesis and the data.

Main Reading Section:

1. Testing of Hypothesis—Proportion: Introduction

Suppose you wish to validate the hypothesis “Most of the vegetarians will face calcium deficiency.” We take a sample (suppose of size 250) from the population consisting of vegetarians from which we will try to identify the number of people having calcium deficiency (suppose 65). Now, 65 out of 250 help us to validate the hypothesis. Here, we are taking the sample proportion to understand the population proportion and, hence, validate the hypothesis.

2. Testing of Hypothesis: One Proportion—Large Sample

Null hypothesis: $H_0: p = p_0$

Test statistic value (large samples):
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Alternative Hypothesis

Rejection Region

$$H_1: p > p_0$$

$$z \geq z_\alpha \text{ (upper-tailed)}$$

$$H_1: p < p_0$$

$$z \leq z_{-\alpha} \text{ (lower-tailed)}$$

$$H_1: p \neq p_0$$

$$\text{either } z \geq z_{\alpha/2} \text{ or } z \leq -z_{\alpha/2} \text{ (two-tailed)}$$

Example: A commonly prescribed drug for relieving nervous tension is believed to be only 60% effective. Experimental results with a new drug administered to a random sample of 100 adults who were suffering from nervous tension show that 70 received relief. Is this sufficient evidence to conclude that the new drug is superior to the one commonly prescribed? Use a 0.05 level of significance.

Solution:

Null hypothesis as $H_0: p = 0.6$

Alternate hypothesis $H_1: p > 0.6$

Level of significance: $\alpha = 0.05$

Critical region: $z > 1.645$

$$\text{Test statistic: } z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.7 - 0.6}{\sqrt{\frac{0.6(0.4)}{100}}} = 2.04$$

Decision: Calculated $z = 2.04 > 1.645$ (table value); hence reject H_0 .

3. Testing of Hypothesis: Two Proportions—Large Sample

Null hypothesis: $H_0: p_1 - p_2 = 0$

Test statistic value (large samples): $z = \frac{\widehat{p}_1 - \widehat{p}_2}{\sqrt{(\widehat{p} \widehat{q}) \left(\frac{1}{m} + \frac{1}{n} \right)}}$

Alternative Hypothesis

Rejection Region for Approximate Level α Test

$$H_1: p_1 - p_2 > 0$$

$$z \geq z_\alpha$$

$$H_1: p_1 - p_2 < 0$$

$$z \leq -z_\alpha$$

$$H_1: p_1 - p_2 \neq 0$$

$$\text{either } z \geq z_{\alpha/2} \text{ or } z \leq -z_{\alpha/2}$$

$$\widehat{p} = \frac{X + Y}{m + n} = \frac{m}{m + n} \cdot \widehat{p}_1 + \frac{n}{m + n} \cdot \widehat{p}_2$$

Example: There is a proposal to change the curriculum in a country as per the global changes and industry demands. The government wants to take the opinion of students and teachers from two universities. In total, 120 out of 200 favored the change in University 1, and 240 out of 500 favored the change in University 2. Would you agree that the proportion of supporters in University 1 is higher than in University 2? Use a 5% level of significance.

Solution:

Null hypothesis as $H_0 : p_1 = p_2$

Alternate hypothesis $H_1 : p_1 > p_2$

Level of significance: $\alpha = 0.05$

Critical region: $z > 1.645$

Test statistic:

$$\widehat{p}_1 = \frac{x_1}{n_1} = \frac{120}{200} = 0.60$$

$$\widehat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 240}{200 + 500} = 0.51$$

$$z = \frac{0.60 - 0.48}{\sqrt{(0.51)(0.49)\left(\frac{1}{200} + \frac{1}{500}\right)}} = 2.9$$

Decision: Calculated $z = 2.9 > 1.645$ (table value); hence reject H_0 .